

Variational Year-Style Translation: A Single-Generator Conditional CycleGAN for Re-Dating Object Imagery

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Abstract

We study *visual re-dating*: rendering an object crop as it would have appeared in a chosen target year while preserving its content. A faithful multi-domain CycleGAN would require a separate generator/discriminator pair for every ordered pair of year buckets, which scales quadratically with the number of buckets. We instead condition a *single* generator on a target year through a *variational year code*: each year is a small Gaussian in a latent space, sampled and mapped to a style vector that modulates the generator via adaptive instance normalization. Year identity is enforced by an auxiliary year classifier in the discriminator, while a Kullback–Leibler term keeps the latent compact and continuous, enabling deterministic targeting, intra-year style variation, and interpolation to unseen in-between years. The model is trained with cycle-consistency, identity, adversarial, classification, and KL objectives. This section details the formulation; experiments are reported in the following section.

1 Method

Our goal is to translate the apparent *date* of an object photograph: given a cropped object and a target year, re-render the object so that its low-level appearance (film response, colour, grain, lighting, and period-specific styling) matches that target year, while preserving object identity and pose. We treat year as a categorical domain and adopt the structure of CycleGAN [Zhu et al., 2017]—two mapping directions tied by cycle consistency and two adversarial signals—but avoid instantiating one network per ordered pair of domains. Instead, following the multi-domain philosophy of StarGAN [Choi et al., 2018], we use a single generator conditioned on a target year, and we make the conditioning *variational* so that years live in a compact, continuous latent space. Figure 1 summarises the training step.

1.1 Problem formulation and notation

Let $\mathcal{X} \subset \mathbb{R}^{3 \times H \times W}$ be the space of object crops, normalised to $[-1, 1]$. Each crop x carries an *origin* year label $y_o \in \{1, \dots, K\}$, where the K labels index contiguous five-year buckets spanning 1930–1999 ($K = 14$). We seek a single conditional generator

$$G : \mathcal{X} \times \mathbb{R}^{d_w} \rightarrow \mathcal{X}, \quad \hat{x} = G(x, w), \quad (1)$$

that re-renders x in a target year encoded by a *style vector* $w \in \mathbb{R}^{d_w}$. Writing w_y for a style vector that represents year y , the translation of x to year y is $G(x, w_y)$; we abbreviate this by $G(x, y)$ when the style network is clear from context. A vanilla two-domain CycleGAN models a single pair $X \leftrightarrow Y$ with two generators and two discriminators; the faithful K -domain generalisation would need $\mathcal{O}(K^2)$ generators. By conditioning one generator on w_y we reduce this to a single network shared across all years.

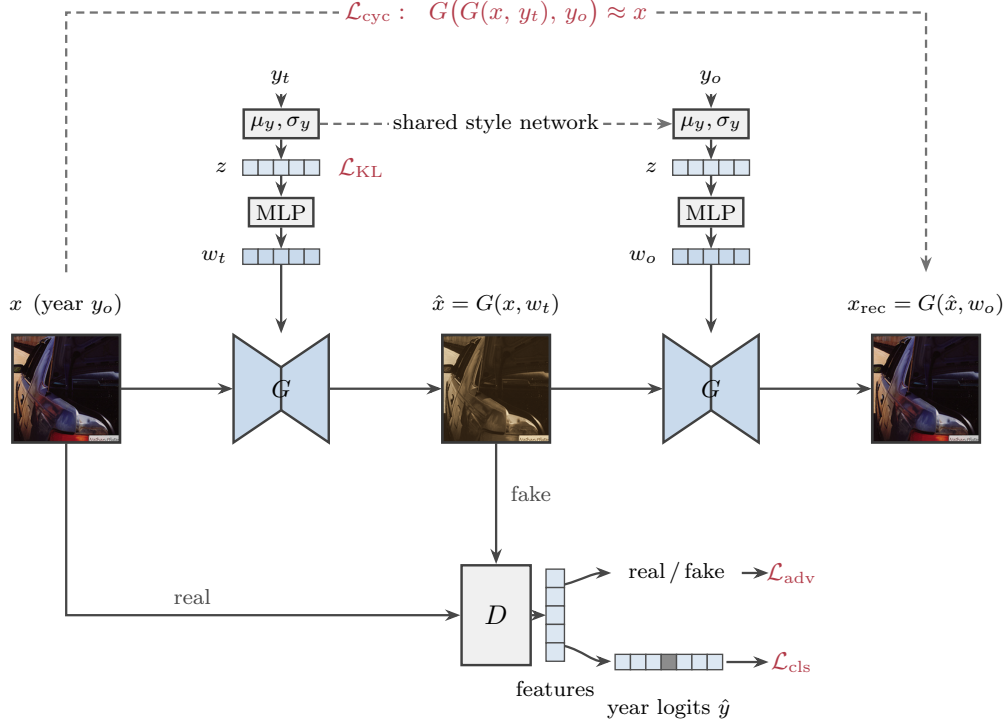


Figure 1: One training step of the year-conditional CycleGAN. A shared style network draws a year code $z \sim \mathcal{N}(\mu_y, \sigma_y^2)$ for the target and origin years and maps it to style vectors w_t, w_o that modulate the shared generator G via adaptive instance normalization. The forward cycle $x \rightarrow \hat{x} = G(x, w_t) \rightarrow x_{\text{rec}} = G(\hat{x}, w_o)$ is constrained by cycle consistency, $G(G(x, y_t), y_o) \approx x$. The discriminator D produces a feature vector from which two heads read a real/fake score (adversarial loss $\mathcal{L}_{\text{adv}}$) and a year posterior (classification loss $\mathcal{L}_{\text{cls}}$); the year code is regularised by $\mathcal{L}_{\text{KL}}$.

1.2 Year as a variational style

We represent each year not as a point but as a compact distribution in a latent space, the probabilistic analogue of a per-year proxy. Concretely, a style network S holds two embedding tables, $\mu \in \mathbb{R}^{K \times d_z}$ and $\log \sigma^2 \in \mathbb{R}^{K \times d_z}$, defining a per-year Gaussian

$$q(z | y) = \mathcal{N}(\mu_y, \text{diag}(\sigma_y^2)). \quad (2)$$

A code is drawn with the reparameterisation trick [Kingma and Welling, 2014],

$$z = \mu_y + \sigma_y \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (3)$$

and mapped to a style vector by a small multilayer perceptron, $w_y = \text{MLP}(z) \in \mathbb{R}^{d_w}$. Setting $\epsilon = \mathbf{0}$ (i.e. $z = \mu_y$) gives a *deterministic* style for year y , while sampling $\epsilon \neq \mathbf{0}$ yields intra-year style variation.

Conditioning the generator. The generator is a residual encoder–decoder [Johnson et al., 2016]: a downsampling stem, a stack of residual blocks, and an upsampling head with reflection padding throughout. The style vector w enters only through the residual blocks, via adaptive instance normalisation [Huang and Belongie, 2017]. For a feature map h with per-channel mean $\mu(h)$ and standard deviation $\sigma(h)$,

$$\text{AdaIN}(h, w) = (\mathbf{1} + \gamma(w)) \odot \frac{h - \mu(h)}{\sigma(h)} + \beta(w), \quad (4)$$

where $(\gamma(w), \beta(w))$ are produced by an affine layer per block and are initialised near zero so that each block starts close to the identity. Because w modulates only normalisation statistics, content is carried by the convolutional features while *year* is injected as a global style—exactly the disentanglement we require for re-dating.

Latent regularisation. To keep each year’s latent compact and the overall space continuous (so that years can be interpolated), we regularise $q(z \mid y)$ towards a standard normal prior with a Kullback–Leibler penalty,

$$\mathcal{L}_{\text{KL}}(y) = \text{KL}(q(z \mid y) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) = \frac{1}{2} \sum_{i=1}^{d_z} (\mu_{y,i}^2 + \sigma_{y,i}^2 - \log \sigma_{y,i}^2 - 1). \quad (5)$$

We use a deliberately small weight on this term: *separation* between years is not provided by the KL (which would, on its own, collapse the years onto the prior) but by the year classifier described next, which forces z to carry year identity. The KL only enforces compactness and continuity.

1.3 Discriminator with an auxiliary year classifier

The discriminator D is a fully convolutional network that maps an image to a spatial feature tensor, from which two linear heads branch (Figure 1, bottom). The first head outputs an unnormalised real/fake field used for the adversarial signal; the second is a *year classifier* that, after spatial pooling, produces a posterior over the K year buckets, $D_{\text{cls}}(y \mid x)$. The classifier plays the role of the StarGAN domain classifier [Choi et al., 2018]: it is what actually *instantiates* a target year, since the generator is rewarded only when its output is classified as the requested year. We use a least-squares adversarial objective [Mao et al., 2017] for stable training,

$$\mathcal{L}_{\text{adv}}^D = \frac{1}{2} \mathbb{E}_x [(D_{\text{src}}(x) - 1)^2] + \frac{1}{2} \mathbb{E}_{\hat{x}} [D_{\text{src}}(\hat{x})^2], \quad \mathcal{L}_{\text{adv}}^G = \frac{1}{2} \mathbb{E}_{\hat{x}} [(D_{\text{src}}(\hat{x}) - 1)^2], \quad (6)$$

where D_{src} denotes the real/fake head. To stabilise the discriminator we draw $\hat{x}$ for the D update from a buffer of previously generated images [Shrivastava et al., 2017].

1.4 Training objective

Given a real crop x with origin year y_o and a target year y_t sampled uniformly at random, we obtain style vectors $w_t = \text{MLP}(z_t)$ and $w_o = \text{MLP}(z_o)$ from Eq. (3), the forward translation $\hat{x} = G(x, w_t)$, the cyclic reconstruction $x_{\text{rec}} = G(\hat{x}, w_o)$, and the identity mapping $G(x, w_o)$.

Year classification. The discriminator’s classifier is trained on *real* images with their true origin year, while the generator is trained so that its output is classified as the *target* year:

$$\mathcal{L}_{\text{cls}}^D = \mathbb{E}_{x, y_o} [-\log D_{\text{cls}}(y_o \mid x)], \quad \mathcal{L}_{\text{cls}}^G = \mathbb{E}_{x, y_t} [-\log D_{\text{cls}}(y_t \mid \hat{x})]. \quad (7)$$

Cycle consistency and identity. Translating to the target year and back to the origin year should recover the input, and translating to the origin year should be a no-op:

$$\mathcal{L}_{\text{cyc}} = \mathbb{E} [\|G(G(x, y_t), y_o) - x\|_1], \quad \mathcal{L}_{\text{idt}} = \mathbb{E} [\|G(x, y_o) - x\|_1]. \quad (8)$$

Both use an ℓ_1 reconstruction penalty. The cycle term is the mechanism that ties content across years; the identity term discourages spurious changes when the requested year already matches the input.

Full objective. The discriminator and the generator (jointly with the style network) are optimised by alternating descent on

$$\mathcal{L}_D = \mathcal{L}_{\text{adv}}^D + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^D, \quad (9)$$

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}}^G + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^G + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{idt}} \mathcal{L}_{\text{idt}} + \lambda_{\text{kl}} (\mathcal{L}_{\text{KL}}(y_t) + \mathcal{L}_{\text{KL}}(y_o)). \quad (10)$$

The style network receives gradients only through $\mathcal{L}_G$: from the adversarial and classification terms (which shape w so that the requested year is realised) and from the KL term (which regularises the latent). Crucially, the classifier gradient is what keeps years distinct, so a small λ_{kl} does not collapse the latent.

1.5 Optimisation and architecture

Both networks use instance normalisation [Ulyanov et al., 2016] and weights initialised from $\mathcal{N}(0, 0.02^2)$. The generator has 64 base channels and 9 residual blocks; the style network uses $d_z = d_w = 64$. We optimise with Adam [Kingma and Ba, 2015] ($\beta_1=0.5, \beta_2=0.999$) under a two time-scale update rule [Heusel et al., 2017], giving the discriminator a faster learning rate than the generator. The learning rate is held constant for the first half of training and then decayed linearly to zero [Zhu et al., 2017]. For sampling we maintain an exponential moving average of the generator and style-network weights. Unless otherwise stated we use $\lambda_{cyc}=10$, $\lambda_{idt}=5$, $\lambda_{cls}=1$, and $\lambda_{kl}=10^{-2}$.

1.6 Inference: targeting, sampling, and interpolation

At test time the variational code supports three modes. *Deterministic targeting* of a year y uses the latent mean, $w_y = \text{MLP}(\mu_y)$, which gives a single canonical rendering. *Stochastic* translation samples $z \sim q(z | y)$ to produce diverse in-year styles for a fixed content. Finally, because the KL term keeps the latent continuous, *unseen or in-between years* are reachable by interpolating the latent means,

$$w_{a \rightarrow b}(t) = \text{MLP}((1-t)\mu_a + t\mu_b), \quad t \in [0, 1], \quad (11)$$

yielding a smooth appearance trajectory between two year buckets without retraining. A single forward pass through G then re-renders any input crop at the chosen point on the year axis.

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